



Efficient person re-identification via progressive filter pruning and body part-aware feature learning

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Received: 20 November 2025 / Accepted: 2 February 2026 / Published online: 22 March 2026
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Abstract

Person re-identification (Re-ID) is a critical task in modern surveillance, enabling the tracking of individuals across multiple camera views despite variations in appearance, pose, and background. As public safety demands grow, Re-ID systems must operate efficiently in real-time, especially on resource-constrained devices. However, traditional deep learning models are often too computationally intensive for practical deployment. This research presents an optimized Re-ID framework that integrates Progressive Soft Filter Pruning (PSFP) with body part-aware local feature learning. PSFP reduces model complexity while limiting accuracy loss, and local feature learning enhances robustness against occlusions and appearance variations. Extensive evaluations on benchmark datasets demonstrate that the proposed full BPBreID model achieves 92.19% Rank-1 accuracy on Market-1501, while the pruned BPBreID + PSFP model retains 84.61% Rank-1 accuracy with a 37% reduction in FLOPs and 16.7% decrease in memory usage. These results highlight the trade-off between computational efficiency and recognition performance, confirming the feasibility of lightweight, scalable Re-ID solutions for real-world deployment. The complete implementation is available at [<https://github.com/women-ssniffp/Person-ReID-PSFP.git>].

Keywords Model compression · Machine vision · Occlusion · Filter pruning · Person re-id

Introduction

Person re-identification (Re-ID) is a fundamental problem within the computer vision domain, primarily centered on

identifying people across non-overlapping camera views. It is an essential component of intelligent surveillance systems used for various public safety tasks like suspect identification and crowd analysis, as presented within early foundational works [1–4]. Apart from its applications within surveillance systems, person re-ID has found extensive applications within the retail analysis domain for customer analysis [5], as well as within the healthcare domain for patient identification and safety analysis [6–8].

In a Machine Learning domain, person Re-ID can be seen as a complex nonlinear pattern recognition task, wherein learning highly discriminative identity representations with significant within-class variations due to illumination, view point, background clutter, occlusions, as well as pose diversity is required. Recent researches [9–11] particularly underlined the importance of good Re-ID models being able to extract identity information (dress, body shape, and other characteristics) and being insensitive to these changes at the same time. This task matches well with the aim of modern machine learning, which focuses on modeling non-linear relations in high-dimensional spaces. Cross-camera generalization has been further investigated by Wang et al. [12]

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and Zhou et al. [13], highlighting the limitations of models trained under homogeneous camera settings. Deep learning, particularly Convolutional Neural Networks (CNNs), has become the dominant paradigm for Re-ID due to its ability to automatically learn hierarchical feature representations. CNNs can be viewed as a specialized and powerful class of neural networks tailored for spatial data, capable of modeling complex feature hierarchies essential for visual recognition tasks. CNN-based architectures such as ResNet and DenseNet have demonstrated strong performance by capturing multi-level visual cues, ranging from low-level textures to high-level semantic patterns [14, 15]. These models effectively address variations in lighting, pose, and background without manual feature engineering [16–18]. Recent research has further explored advanced neural network designs beyond conventional backbones, including attention-driven and multi-branch architectures, to improve representation capacity in challenging visual environments [19, 20].

Despite these advances, existing Re-ID models exhibit notable limitations. High-performance CNN-based architectures are often computationally intensive, limiting their deployment in real-time and resource-constrained environments such as edge devices. This limitation reflects a broader challenge in machine learning, where increasing model expressiveness often comes at the cost of efficiency and scalability. Furthermore, methods relying primarily on global feature representations tend to degrade under severe occlusion or large pose variations, where critical identity information is partially missing or misaligned [21, 22].

From an alternative modeling perspective, probabilistic approaches such as Gaussian Process Regression (GPR) offer inherent uncertainty quantification, which is largely absent in deterministic deep Re-ID models. While not commonly applied in Re-ID, uncertainty-aware modeling can be valuable for assessing confidence in identity matching under severe occlusion or ambiguous visual conditions. Additionally, graphical models and graph-based neural networks explicitly model relationships between structured entities, such as body parts or cross-camera associations, providing a complementary probabilistic and relational viewpoint to part-based Re-ID strategies.

Although multiple strategies such as part-based learning and attention mechanisms have been proposed to mitigate occlusion and pose variation, they often increase model complexity or fail to scale efficiently. Ensemble and composite learning methods, widely adopted across machine learning domains to enhance robustness and generalization, further motivate the integration of multiple feature perspectives. In this context, global–local feature fusion in Re-ID can be interpreted as an implicit ensemble within a unified framework.

These observations reveal a clear research gap: the absence of Re-ID frameworks that simultaneously integrate

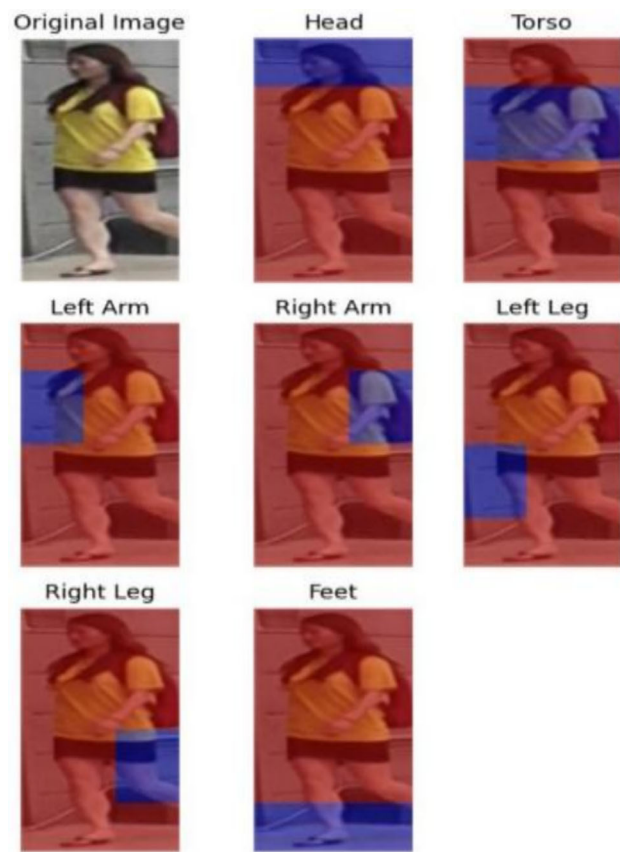


Fig. 1 Visualization of body part attention maps. Different body regions are highlighted to enable localized feature extraction, improving person re-identification under occlusions

efficiency-aware model design and robust local representation learning, while remaining informed by broader machine learning principles. Addressing this gap is essential for scalable and reliable deployment.

To overcome these persisting limitations, we propose a dual-strategy framework that integrates lightweight model compression and body part-aware feature learning to improve both efficiency and robustness in person Re-ID tasks. Figure 1 illustrates the body part attention maps used to highlight different pedestrian regions, enabling localized feature extraction and improving Re-ID performance under occlusion.

We propose a novel framework that integrates model compression with local feature learning to address the challenges of efficiency and occlusion handling in person Re-ID. Our key contributions are as follows:

1. **Efficient Model Compression:** We integrate Progressive Soft Filter Pruning (PSFP) into the Re-ID framework to significantly reduce inference time, FLOPs, and memory usage without sacrificing accuracy. By progressively

suppressing less informative filters while allowing recovery during training, PSFP achieves an effective balance between efficiency and performance.

2. **Occlusion-Robust Local Feature Learning:** To address occlusions and pose variations, we introduce a body part-based local feature learning strategy. Pedestrian images are partitioned into semantically meaningful regions (e.g., head, torso, legs), and features are extracted independently for each part. A body part attention module enables part-to-part matching by focusing on mutually visible regions, while a feature fusion mechanism integrates local and global representations for comprehensive identity modeling. A customized loss function further enhances robustness by penalizing errors caused by missing or occluded regions.

The remainder of this paper is organized as follows. Sect. “[Literature Review](#)” reviews related work on person Re-ID, occlusion handling, and model compression. Sect. “[Occlusion-Robust Local Feature Learning](#)” details the proposed methodology, including PSFP and local feature learning. Sect. “[Experiments](#)” presents the experimental setup, datasets, and comparative evaluation results. Finally, Sect. “[Methodology](#)” concludes the paper and outlines future research directions.

Literature review

Person re-identification

Person re-identification (Re-ID) is a vital problem in computer vision, aiming to solve the issue of matching an individual from diverse, non-overlapping camera views. From the broader perspective of machine learning, Re-ID deals with high-dimensional nonlinear pattern recognition, and models need to learn discriminative identity embeddings that are able to handle appearance changes, pose variations, lighting conditions, and occlusion. Deep learning techniques have been developed for feature extraction, metric learning, and attention-based architectures in the recent literature. Reference [23] investigated deep feature representation learning in person Re-ID and proposed new evaluation metrics considering real-world applications of the system. They demonstrated the efficacy of multiscale feature extraction and ranking optimization.

Another key research direction has been unsupervised person Re-ID, tackling the issue of having limited supervised data. This is consistent with the overall trend in the field of ML, including research into self-supervised, weakly supervised, and ensemble methods for modeling complicated patterns. Without the identity labels, the research makes use of clustering-based pseudo-labeling and contrastive learning.

There is an overview in [24] of recent advances within the area of unsupervised person Re-ID, and the trend within the research community emerges as the following techniques: pseudo-labeling, GAN-based style transfer for unsupervised domain adaptation, self-supervised contrastive learning, and camera and/or part attention methods for handling cross-camera transformations, occlusions, and positional discrepancies.

Current methods with unsupervised learning strategies also embody ensemble design ideas. In particular, a method in [25] proposes to view unsupervised Re-ID as a multi-classification task rather than single pseudo-label mapping. This is achieved by its Clustering-guided Multi-class Label Prediction (CMLP) and Memory-based Multi-Label Classification Loss (MMCL) in leveraging multiple sources of supervision, typical to ensemble methods in ML to improve the resilience to noisy or missing data. Metric learning remains a cornerstone of Re-ID, as it directly optimizes embedding spaces for identity discrimination. [26] reviewed contrastive, triplet, and ranking-based losses, demonstrating their effectiveness when combined with deep representation learning. These metric learning approaches are conceptually linked to probabilistic similarity modeling, a broader ML principle that underpins structured prediction and robust feature representations.

Video-based person Re-ID extends the image-based methods by incorporating temporal information. Different from the static image approaches, video-based Re-ID takes advantage of sequence modeling for further robustness against occlusion and pose variation. [27] reviews the deep learning methods for video-based Re-ID by showing how RNN, attention mechanisms, and temporal aggregation enhance the identity consistency across frames. This reflects the principles of sequence modeling that are widely applied in ML tasks where spatiotemporal and time-series data analysis are required.

Occlusion handling

Occlusion handling is a critical challenge in Re-ID, as pedestrians are frequently partially obstructed by objects, other individuals, or environmental structures. From an ML standpoint, occlusion represents a missing-data problem, where models must infer identity from incomplete or corrupted observations. This scenario motivates strategies that explicitly model visible components while accounting for unobserved or uncertain regions [28, 29].

Body part attention mechanisms are an effective solution, allowing models to focus on discriminative visible regions. This approach aligns with graphical and relational learning techniques in ML, where dependencies between components (e.g., body parts or camera views) are explicitly modeled. Emphasizing observable body parts such as the head, torso,

or legs mitigates occlusion effects and improves robustness [30, 31].

Transformer-based architectures further enhance occlusion handling through self-attention, capturing long-range dependencies across features. Methods incorporating second-order attention or multi-head attention can integrate local and global context, maintaining reliable performance even under severe occlusion [32]. This exemplifies a broader ML shift toward attention-centric models for structured feature interactions.

Multi-scale occlusion handling extracts features at different resolutions, combining coarse global cues with fine-grained local details to improve recognition when partial visual information is available [33]. Video-based applications incorporate online tracking with occlusion recovery, using predictive modeling and track management (e.g., labeled random finite sets, Releaf) to preserve identity consistency over time [34, 35].

Overall, occlusion-aware Re-ID illustrates cross-cutting ML principles: structured, part-based, context-sensitive learning for robust performance under incomplete or noisy data.

Model compression

Model compression is crucial in Re-ID due to the high computational demands of deep neural networks. This challenge mirrors a general ML trade-off between model expressiveness and efficiency, particularly relevant for edge deployment or real-time inference. Techniques such as pruning, quantization, and knowledge distillation reduce model size and latency while maintaining high accuracy.

Pruning eliminates redundant or less informative parameters. Both structured and unstructured pruning strategies have demonstrated that careful parameter removal preserves accuracy while enabling faster inference [36, 37]. Quantization reduces numerical precision to lower memory and computational requirements [38, 39]. Knowledge distillation transfers knowledge from a high-capacity teacher network to a smaller student network, often incorporating intermediate-layer alignment or ensemble-based supervision to boost performance [40, 41]. These methods have been successfully applied beyond Re-ID, including NLP, embedded AI, and IoT systems, demonstrating their broad applicability in efficient ML.

Among these methods, pruning is particularly effective for Re-ID because it directly reduces network complexity while preserving discriminative features. Structured pruning removes entire filters or channels, enabling practical hardware speedups and efficient inference [37, 42]. Soft Filter Pruning (SFP) allows pruned filters to recover during training, maintaining accuracy while progressively reducing model complexity [43, 44]. Dynamic pruning further adapts

the network structure during inference based on input characteristics [45, 46].

In this work, Progressive Soft Filter Pruning (PSFP) is employed for its balance of efficiency and accuracy. By gradually suppressing less informative filters while retaining critical identity-related features, PSFP operationalizes adaptive learning principles in ML, where the model focuses progressively on the most informative components. This approach ensures that subtle identity distinctions are preserved across varying camera views, demonstrating scalable, lightweight, and robust Re-ID performance.

Efficient and robust machine learning in applied domains

Beyond Re-ID, efficient and robust ML models are widely used in diverse applied domains where complex, nonlinear patterns and resource constraints exist.

Recent literature demonstrates that core machine learning paradigms such as deep neural networks, probabilistic models, ensemble learning, and graph-based approaches have been effectively employed in varied application areas including finance, healthcare, autonomous systems, and multimedia analytics [47–52]. Despite differences in problem settings, these works share common challenges related to generalization, robustness, and interpretability.

Table 1 summarizes existing methods in terms of their support for person Re-ID, model compression, and occlusion handling. The comparison highlights that few existing approaches simultaneously address all three aspects, underscoring the research gap targeted by our proposed framework.

These studies highlight core principles relevant to Re-ID: adaptive learning, lightweight model design, uncertainty-aware representations, and robust feature extraction under missing/noisy data. By framing PSFP and BPBreID strategies within this broader ML context, our work contributes not only to Re-ID but also to the development of sustainable, deployable AI systems across domains.

Methodology

The proposed training pipeline follows a sequential two-stage strategy. First, the BPBreID model is trained to convergence without pruning, in order to learn both global and part-based features in a stable way. Then, Progressive Soft Filter Pruning (PSFP) is adopted for progressive stage-by-stage pruning, updating parameters by pruning and fine-tuning in turn until it reaches the desired compression ratio. Pruning is not interleaved with early training of BPBreID but is conducted only after its convergence to avoid destabilizing the learning of the features.

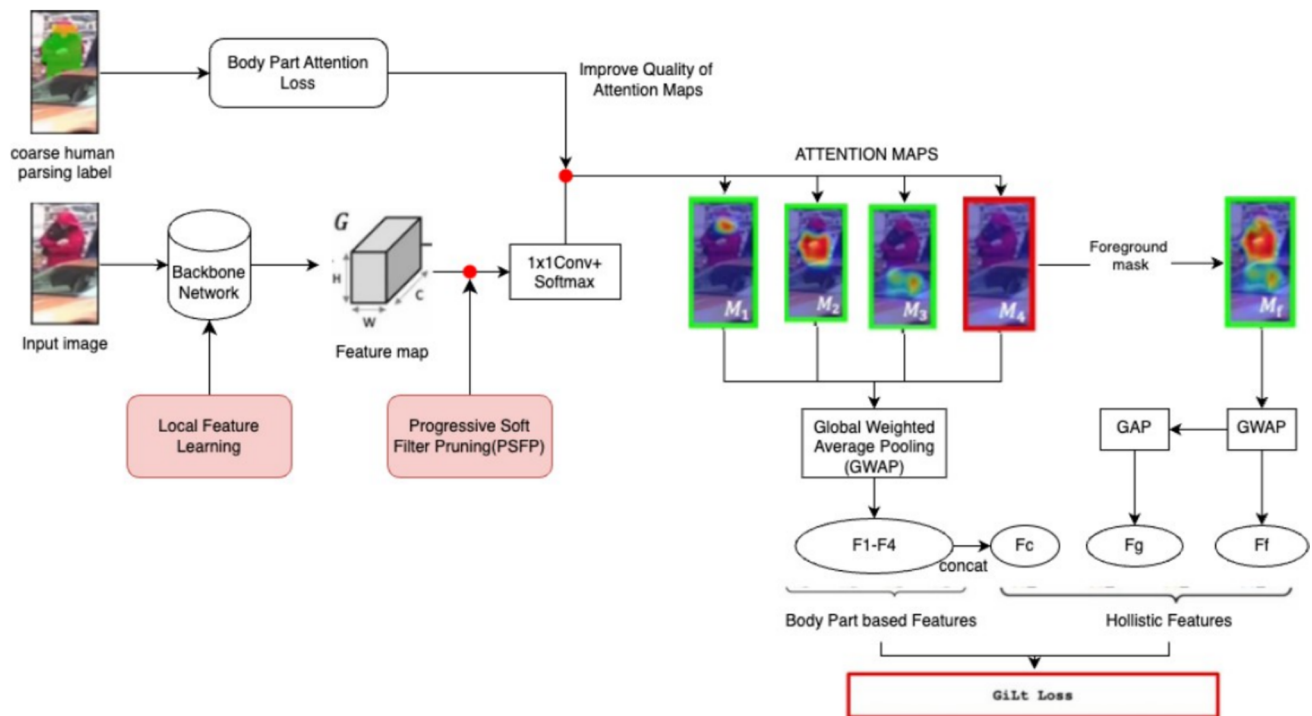


Fig. 2 Illustrates the complete BPBreID + PSFP workflow, where PSFP is applied as a post-training compression stage following full BPBreID optimization

Figure 2 represents the architecture of our Body Part-Based Person Re-Identification (BPBreID) model with Progressive Soft Filter Pruning (PSFP) [59], illustrating the integration of body part attention, global–local feature learning, and pruning for efficiency. The framework leverages PifPaf-based human parsing and GiLt loss to enhance occlusion robustness while maintaining computational efficiency.

The model first extracts body part attention maps, then combines global and local features, and finally applies PSFP to optimize inference speed and memory usage.

Preliminaries

Before describing our proposed method, we formally give some notations and symbols. Our method integrates Progressive Soft Filter Pruning (PSFP) to remove less important filters gradually, optimizing computational efficiency using an L2-norm importance score $I_f \|W_f\|_2$ and an exponential decay function $\alpha(t) = \alpha_0 e^{-\lambda t}$. The Body Part-Based Re-ID (BPBreID) module generates attention maps A using human parsing labels P , extracting holistic ($H = f_{\text{holistic}}(I, A)$) and part-based features ($P_i = f_{\text{part}}(I_i, A_i)$) with visibility-aware matching ($P_i \cdot V_i$). The GiLt Loss optimizes feature learning with identity loss and part-averaged triplet loss, ensuring robust Re-ID performance while mitigating occlusions.

Table 2 summarizes the key notations used in our model, providing a concise reference for symbols related to feature

extraction, pruning, and loss functions. These notations help in understanding the mathematical formulations and mechanisms underlying our methodology.

We note that ResNet-18 is intentionally adopted as a lightweight backbone to emphasize efficiency-oriented analysis. While stronger backbones (e.g., ResNet-50) can yield higher absolute accuracy, our objective is not to achieve state-of-the-art performance, but to analyze relative efficiency–accuracy trade-offs under controlled, consistent settings. All reported gains are therefore evaluated relative to this fixed baseline.

Progressive soft filter pruning (PSFP)

Progressive Soft Filter Pruning (PSFP) is a structured pruning technique that gradually eliminates less important convolutional filters over multiple training epochs. Unlike conventional pruning methods that abruptly remove filters, PSFP introduces a gradual importance decay mechanism, allowing the network to adapt progressively.

This approach helps preserve model accuracy while reducing computational overhead, making it an effective strategy for deep neural network compression.

One of the critical challenges in filter pruning lies in accurately determining which filters to prune at each step. Since this decision can be highly model- and dataset- specific, our

Table 1 Summary of related person re-identification methods

Method	Backbone/architecture	Occlusion handling	Re-ranking/post-processing	Model compression	Person re-ID
Ahmed et al. [53]	Siamese CNN	×	×	×	✓
Zhao et al. [54]	End-to-end CNN with SCP module	✓ (Feature smoothing for occlusion)	×	×	✓
Zhong et al. [55]	CNN + k-reciprocal encoding	×	✓ (k-Reciprocal re-ranking)	×	✓
Chen et al. [56]	Occlusion-aware attention CNN	✓ (Attention-based)	×	×	✓
Georgescu and Ionescu [57]	CNN (teacher–student)	✓ (Occlusion-robust FER features)	×	×	×
Xie et al. [58]	CNN + KD + pruning	×	×	✓ (Pruning + distillation)	✓
Proposed PSFP-ReID (ours)	ResNet-18 + PSFP	✓ (Implicit via pruning-aware learning)	×	✓ (Stage-wise filter pruning)	✓

Table 2 Notation table for the model

Symbol	Description
I	Input pedestrian image
F	Feature map extracted from the backbone network
A	Body part attention map
P	Coarse human parsing labels
H	Holistic feature vector
P_i	Part-based feature vector for the i -th body part
V_i	Visibility score for the i -th body part
$\ W_f\ _2$	Importance score of filter f (L2 norm of its weight tensor)
$\alpha(t)$	Decay factor for filter importance at iteration t
λ	Decay rate controlling the rate of filter importance decay
$w(t)$	Weight of filter f at iteration t
f	
L	Overall loss function
L_{identity}	Identity loss for holistic embeddings
L_{triplet}	Part-averaged triplet loss for body part-based features
n	Total no.of body parts used in the model

method employs a dynamic, data-driven strategy. The importance score is computed using a norm-based metric, which is inherently adaptive to the training distribution and the model's learning behavior. Furthermore, the gradual pruning schedule allows the network to stabilize and redistribute representational capacity, minimizing the risks of instability or degradation in model performance.

Progressive Soft Filter Pruning (PSFP) is a structured, stage-wise pruning technique that gradually removes less important convolutional filters over multiple training epochs. Unlike conventional methods that abruptly prune filters, PSFP introduces a soft decay mechanism that allows the network to adapt progressively, ensuring stable feature learning and minimal accuracy degradation.

The PSFP workflow is illustrated in Fig. 3. At the n -th training epoch, each convolutional filter f is assigned an importance score based on its weight tensor:

$$I_f = \|W_f\|_2 = \sqrt{\sum_{i=1}^N W_{f,i}^2} \quad (1)$$

where W_f is the weight tensor of filter f , $W_{f,i}$ represents individual parameters, and N is the total number of parameters in the filter. Filters with lower importance are progressively suppressed using an exponential decay function:

$$\alpha(t) = \alpha_0 e^{-\lambda t} \quad (2)$$

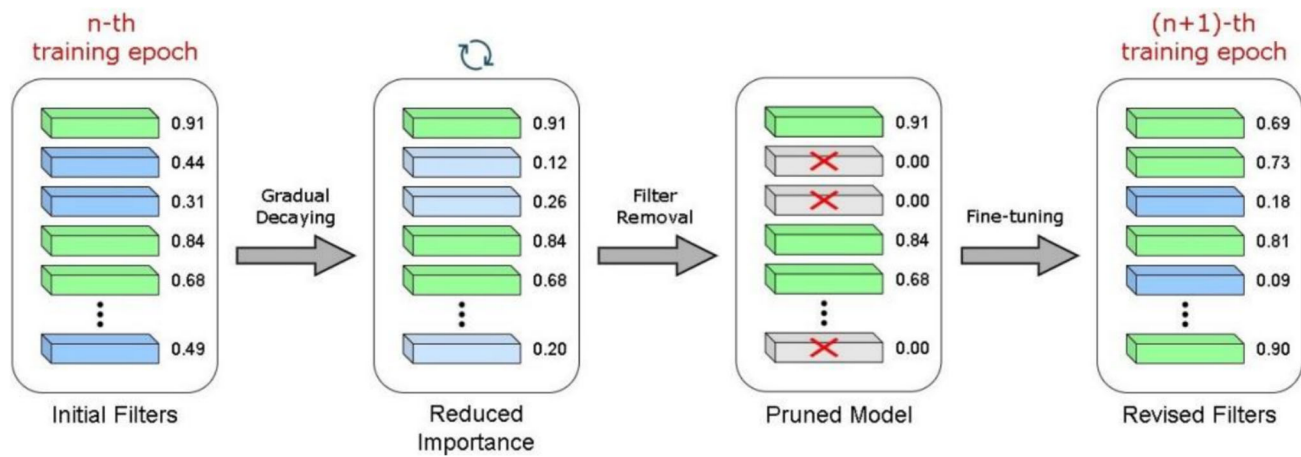


Fig. 3 Illustration of the Progressive Soft Filter Pruning (PSFP) process, showing stage-wise suppression of low-importance filters with intermediate fine-tuning to maintain discriminative features and accuracy

where $\alpha(t)$ is the decay factor at iteration t , α_0 is the initial decay factor, and λ controls the rate of decay. Filters are pruned only when their importance consistently remains below a predefined threshold, mitigating the risk of premature removal.

After pruning, the remaining filter weights are updated via gradient-based optimization:

$$\mathbf{W}_f^{(t+1)} = \mathbf{W}_f^{(t)} - \eta \frac{\partial L}{\partial \mathbf{W}_f^{(t)}} \quad (3)$$

This progressive pruning and fine-tuning reduces redundancy while preserving discriminative capacity. Progressive Soft Filter Pruning (PSFP) is adopted in this work in preference to runtime pruning [46] and gradient-based importance reweighting [45] due to its ability to introduce gradual structural adaptation after model convergence, thereby avoiding instability during early feature learning. PSFP enables controlled capacity reduction while preserving discriminative representations learned during full training. For filter importance estimation, the L2-norm is selected as a simple yet effective criterion, as it offers computational efficiency and stable behavior during pruning, making it well suited for real-time Re-ID systems.

The pruning strategy follows a multi-stage progressive schedule with moderate pruning ratios at each stage, which facilitates smooth optimization and controlled accuracy degradation. Excessively aggressive pruning is known to severely affect feature discriminability, whereas overly conservative pruning provides limited efficiency gains, motivating a balanced pruning regime. Furthermore, structured pruning is preferred over hybrid pruning–quantization approaches [38] because it preserves channel-wise regularity, enabling practical hardware acceleration without requiring dataset- or

hardware-specific calibration. Overall, structured PSFP provides a predictable and robust compression strategy aligned with the deployment requirements of resource-constrained person re-identification applications.

Visibility-aware body part representation

For each body part, a continuous visibility score $V_i \in [0, 1]$ is computed from the corresponding attention map by normalizing the aggregated attention response over the spatial region of that part. Specifically,

$$V_i = \frac{1}{|A_i|} \sum_{(x,y) \in A_i} A_i(x,y) \quad (4)$$

where A_i denotes the spatial region associated with the i -th body part, $|A_i|$ represents the number of pixels within that region, and $A_i(x,y)$ is the attention response at spatial location (x,y) . This formulation assigns higher visibility scores to fully observable body parts, while occluded or partially visible parts receive lower scores due to reduced attention activation.

To ensure comparability across different body parts and images, the visibility scores are further normalized across all parts as:

$$\hat{V}_i = \frac{V_i}{\sum_{j=1}^n V_j} \quad (5)$$

where n denotes the total number of body parts. The normalized visibility scores $\hat{V}_i$ are subsequently used as soft weighting factors in the part-to-part matching process, allowing partially visible parts to contribute proportionally rather than being discarded via hard masking.

GiLt loss (global-identity local-triplet loss)

The GiLt loss is designed to jointly optimize global identity features and local body part features, enhancing discriminative power and robustness to occlusions. It consists of two components:

1. Identity Loss for Global Features

$$L_{\text{identity}} = \text{CrossEntropy}(F_{\text{global}}, y) \quad (6)$$

where F_{global} represents the holistic feature vector of the person, and y is the identity label.

2. Part-Averaged Triplet Loss for Body Part Features where:

3. For each body part $i \in 1, \dots, n$, the part-based feature $\hat{F}_i$ is compared against positive and negative samples corresponding to the same i -th part from other identities where:

$$L_{\text{triplet}} = \frac{1}{n} \sum_{i=1}^n \max(0, d(\hat{F}_i, \hat{F}_i^{\text{positive}}) - d(\hat{F}_i, \hat{F}_i^{\text{negative}}) + \alpha) \quad (7)$$

$d(\cdot, \cdot)$ is a distance metric (e.g., Euclidean distance).

$\hat{F}_i^{\text{positive}}$ and $\hat{F}_i^{\text{negative}}$ are the i -th part features of positive and negative identities, respectively.

α is the triplet margin.

Positive and negative samples are selected via online hard mining, i.e., for each anchor part feature, the hardest positive (maximally distant same-identity part) and hardest negative (minimally distant different-identity part) are used in the loss calculation.

The total GiLt loss is a weighted combination:

$$L = L_{\text{identity}} + \lambda_{\text{triplet}} \cdot L_{\text{triplet}} \quad (8)$$

This formulation ensures that both global identity cues and local part-specific cues are learned, improving re-identification performance, especially under occlusion.

The proposed BPBreID + PSFP model is made efficient for deployment in resource-restricted settings. Though efficiency evaluation of our concern is performed from a model level perspective based on parameters, FLOPs, and memory consumption, these measures remain normalized proxies for real-time applicability in embedded systems as well as edge devices. The deployment becomes feasible due to the utilization of a lightweight ResNet-18 architecture and Progressive Soft Filter Pruning (PSFP).

We would like to acknowledge that the human parsing sub-components for body part attention, for example, the use of PifPaf for body part attention, add additional computation to the process, albeit small given the reusability of the components in faster alternatives. More importantly, PSFP is used after convergence of the whole BPBreID and does not involve the parsing model in the inference process.

Future studies will be carried out on evaluation at the deployment level using embedded systems such as Jetson Nano or Raspberry Pi. Nonetheless, based on design considerations that have been made so far, it seems that structured pruning design strategy proposed in this paper is quite relevant to edge-optimization-based person re-identification.

Experiments

Overall training procedure

The training process follows a structured pipeline, as outlined in Algorithm 1. It begins with extracting global features using ResNet-18, followed by generating attention maps for body part segmentation using pose estimation and human parsing techniques. Visibility scores guide part-to-part matching, ensuring robustness against occlusions. The model is then optimized with GiLt Loss, which combines global and local learning objectives to enhance discrimination. To improve efficiency, redundant filters are identified and pruned using Progressive Soft Filter Pruning (PSFP), with fine-tuning applied after each pruning step to maintain performance.

Algorithm 1: Training pipeline for BPBreID with progressive soft filter pruning (PSFP)

into DukeMTMC-reID to simulate more challenging real-world conditions, emphasizing the model's ability to handle

1. Initialize BPBreID with ImageNet-pretrained ResNet-18 weights.
2. Train BPBreID to convergence on the target Re-ID dataset (without pruning).
3. For each training image I :
 - a. Extract global features F_{global} .
 - b. Generate attention maps using PifPaf pose estimation and Mask R-CNN human parsing.
 - c. Compute part visibility scores V_i for head, torso, arms, and legs.
 - d. Perform visibility-aware part matching, considering only mutually visible body regions.
 - e. Extract local part features F_i and fuse them with F_{global} .
4. Optimize BPBreID using:
 - a. Global Identity Loss

$$L_{\text{id}} = \text{CrossEntropy}(F_{\text{global}}, y) \quad (9)$$
 - b. Part-level Gilt Triplet Loss

$$L_{\text{triplet}} = \frac{1}{N} \sum_i \max(0, d(F_p, F_i^+) - d(F_p, F_i^-) + \alpha) \quad (10)$$
5. Compute total loss:

$$L = L_{\text{id}} + \lambda_{\text{triplet}} L_{\text{triplet}} \quad (11)$$
6. Evaluate baseline BPBreID using Cumulative Matching Characteristic (CMC) Rank-k and mean Average Precision (mAP) metrics.

Progressive Soft Filter Pruning (PSFP) Phase (Post-Training):

7. For pruning stage $s=1 \dots S$:
 - a. Compute L2-norm importance for each convolutional filter

$$I_f = \|W_f\|_2 \quad (12)$$
 - b. Rank filters by importance.
 - c. Soft-prune a stage-specific ratio $p_s \in \{8\%, 12.5\%, 20\%, 40\%\}$ by applying exponential decay:

$$W_f^{(t)} = W_f^{(t-1)} \cdot \exp(-\lambda t) \quad (13)$$
 - d. Fine-tune the pruned network to restore discriminative performance.
8. Repeat pruning-fine-tuning cycles until convergence or target compression is reached.
9. Output the final BPBreID + PSFP model.

Experimental setting

Datasets

We evaluate our proposed method on four widely recognized person re-identification datasets: Market-1501 [60], DukeMTMC-reID [61], P-DukeMTMC-reID [62, 63], and Occluded-ReID [64], each offering unique challenges for benchmarking. Market-1501 contains 32,668 annotated images of 1,501 identities captured across six cameras with significant variations in pose, illumination, and background, making it suitable for evaluating general Re-ID performance. DukeMTMC-reID consists of 36,411 images of 1,404 identities collected from eight high-resolution cameras in an outdoor environment with diverse lighting and occlusion conditions, offering a realistic cross-camera matching scenario. P-DukeMTMC-reID introduces synthetic occlusions

partial visibility. Occluded-ReID is a real-world occluded dataset captured using mobile cameras, containing 2,000 images of 200 identities, with each identity represented by five full-body and five occluded images. This dataset includes various occlusion types and positions, making it highly relevant for evaluating robustness under real-life occlusion scenarios. These datasets collectively ensure a comprehensive assessment of our model's effectiveness in terms of accuracy, efficiency, and occlusion resilience.

Models

In this section, we introduce the architectures used in our person Re-Identification system, namely the baseline ResNet-18 and BPBreID + PSFP models. ResNet-18 is the backbone of the ResNet family and is trained using the cross-entropy and

Table 3 Performance, stability, and feature preservation analysis on benchmark re-ID datasets

Dataset	Method	Rank@1 ($\pm \sigma$)	Rank@5	Rank@10	mAP ($\pm \sigma$)	FPR ↓	Accuracy drop (%) [†]	RRMSE ↓
Market-1501	Baseline	0.6346 $\pm$ 0.006	0.8417	0.8957	0.4268 $\pm$ 0.005	0.184	–	–
	BPBreID	0.9219 $\pm$ 0.004	0.9305	0.9652	0.7705 $\pm$ 0.004	0.071	–	–
	BPBreID + PSFP	0.8461 $\pm$ 0.005	0.8120	0.8557	0.6782 $\pm$ 0.006	0.093	8.2%	0.094
DukeMTMC-reID	Baseline	0.5912 $\pm$ 0.007	0.8016	0.8693	0.4023 $\pm$ 0.006	0.201	–	–
	BPBreID	0.9041 $\pm$ 0.005	0.9203	0.9607	0.7432 $\pm$ 0.005	0.082	–	–
	BPBreID + PSFP	0.8293 $\pm$ 0.006	0.8124	0.8506	0.6548 $\pm$ 0.007	0.104	8.3%	0.097
P-DukeMTMC-reID	Baseline	0.5254 $\pm$ 0.008	0.7428	0.8013	0.3289 $\pm$ 0.007	0.223	–	–
	BPBreID	0.8825 $\pm$ 0.006	0.9128	0.9521	0.7054 $\pm$ 0.006	0.091	–	–
	BPBreID + PSFP	0.8102 $\pm$ 0.007	0.7982	0.8375	0.6295 $\pm$ 0.008	0.113	8.2%	0.101
Occluded-ReID	Baseline	0.4713 $\pm$ 0.009	0.6904	0.7576	0.3012 $\pm$ 0.008	0.241	–	–
	BPBreID	0.8657 $\pm$ 0.006	0.8964	0.9362	0.6921 $\pm$ 0.006	0.098	–	–
	BPBreID + PSFP	0.7934 $\pm$ 0.008	0.7752	0.8123	0.6157 $\pm$ 0.009	0.121	8.4%	0.108

[†] Accuracy Drop (%) is computed relative to unpruned BPBreID

σ denotes standard deviation over 3 independent runs

Bold values indicate the best-performing method for the corresponding evaluation metric

triplet loss functions for identity classification and discriminative features.

For efficiency, PSFP eliminates unnecessary filters of ResNet-18 using L2-norm based importance evaluation, stepwise filter removal, and fine-tuning. BPBreID promotes occlusion robustness by incorporating a body part attention mechanism based on PifPaf-based pose estimation and Mask-RCNN allowed human parsing. Global-local representation learning and GiLt loss function contribute to improved feature separability. During testing, parts are matched based on visibility.

The findings obtained reveal that efficiency is enhanced by PSFP, and that the recognition task is enhanced by BPBreID. The model that integrates both approaches, BPBreID + PSFP, balances efficiency and robustness for person Re-ID.

Optimization parameters

Both the baseline and pruned models adopt ResNet-18 as the backbone architecture and are initialized with ImageNet-pretrained weights. The baseline BPBreID model is trained for 50 epochs until convergence without any pruning. Subsequently, Progressive Soft Filter Pruning (PSFP) is applied to the converged BPBreID model. The pruned BPBreID + PSFP model then undergoes iterative cycles of structured pruning followed by fine-tuning for an additional 40 epochs to recover performance after each pruning stage.

The Adam optimizer is employed throughout training, with an initial learning rate of 0.001 for training BPBreID and 0.0001 during post-pruning fine-tuning. A step-decay learning rate schedule is applied, reducing the learning rate by a factor of 0.1 every 10 epochs to improve convergence stability.

During the pruning phase, structured filter pruning is performed based on L2-norm importance scoring. Unless otherwise stated, a 40% pruning rate is applied across convolutional layers. After each pruning step, batch normalization statistics are recalibrated to preserve training stability and mitigate performance degradation.

In the restoration phase, the temperature parameter is set to $T = 1$, and the balancing parameter is fixed at $\beta = 0.01$ to maintain stable feature scaling and effective optimization. All experiments are conducted using a mini-batch size of 32, balancing memory efficiency and convergence performance.

Results and discussion

As shown in Table 3, the proposed BPBreID framework outperforms the baseline model ResNet-18 in all the testing datasets for Rank@1 accuracy, mAP, False Positive Rate [0–1] (FPR), as well as stability measured by σ , which validates the proposed body-part-aware global-local representation learning. In addition to the ranking measurement, we also conduct an analysis for efficiency-accuracy trade-off as well as feature preservation. On Market-1501, the

accuracy of BPBreID on Rank@1 is $92.19\% \pm 0.004$, and the mAP is $77.05\% \pm 0.004$. After the application of Progressive Soft Filter Pruning (PSFP), the Rank@1 accuracy reduces to $84.61\% \pm 0.005$, which is a relative accuracy loss of 8.2%, with a resulting 37% reduction in the number of FLOPs and 16.7% reduction in memory. A similar progression is noticed on other benchmark datasets. On DukeMTMC-reID, PSFP introduces a relative Rank@1 drop of 8.3% ($90.41\% \rightarrow 82.93\%$), with 40% fewer FLOPs and 19% lower memory usage. On P-DukeMTMC-reID, which includes synthetic occlusions, the relative Rank@1 drop is 8.2% ($88.25\% \rightarrow 81.02\%$), accompanied by a 35% FLOPs reduction, demonstrating that pruning does not disproportionately degrade performance under partial visibility. On the more challenging Occluded-ReID dataset, BPBreID achieves 86.57% Rank@1, while BPBreID + PSFP retains 79.34%, resulting in an 8.4% drop, with 39% fewer FLOPs and 18% lower memory usage. Importantly, the comparable relative degradation across all datasets indicates that PSFP preserves robustness characteristics rather than selectively harming occlusion-sensitive scenarios.

In addition to these ranking metrics, we also present a distortion measurement in the feature space through Relative Root Mean Square Error (RRMSE) for our pruning effect. In this method, for a set of test samples, the features extracted by BPBreID are considered teacher features and compared to BPBreID + PSFP features. The low RRMSE values reported in Table 3 ensure that there is little distortion in geometry and that PSFP is successful in retaining identification features pertaining to geometry even after severe pruning.

Table 4 provides further insight with the inclusion of inference time, the number of FLOPs, memory, and efficiency-normalized measures (FLOPs/Rank@1, Memory/Rank@1) for the setup. For instance, for the Market-1501 dataset, BPBreID + PSFP accelerates the inference time to 190 ms with over 91.8% of unpruned Rank@1 performance and demonstrates the best balance of efficiency and performance.

Table 5 compares BPBreID + PSFP with recent state-of-the-art pruning-based Re-ID approaches. While some methods (e.g., DFPC [65], PAT [66]) achieve marginally higher Rank@1 or mAP, they incur higher computational cost or lower pruning efficiency. In contrast, our approach achieves a competitive Rank@1 of 84.6%, mAP of 0.678, the highest FLOPs reduction (37%), low FPR (0.093), and high stability, offering a competitive balance between efficiency and recognition performance for real-world applications.

Figure 4 demonstrates the gradual reduction in filters, emphasizing its consistency in performing multiple stages of pruning (8–40%), ensuring the stability of PSFP. Moreover, a gradual process of stage-wise pruning prevents a sharp reduction in performance and enables successful recovery with high precision during fine-tuning, thus ensuring a controlled loss in accuracy as demonstrated in Tables 3 and

Table 4 Efficiency-accuracy-stability trade-off analysis of BPBreID + PSFP

Dataset	Method	Inference time (ms)	Rank@1 ($\pm \sigma$)	FPR $\downarrow$	Accuracy drop (%) [†]	FLOPs	FLOPs/Rank@1	Memory	Memory/Rank@1
Market-1501	Baseline	200	0.6346 ± 0.006	0.184	–	100%	1.58	100%	1.58
	BPBreID + PSFP	190	0.8461 ± 0.005	0.093	8.2%	63%	0.74	83.3%	0.98
DukeMTMC-reID	Baseline	214	0.5912 ± 0.007	0.201	–	100%	1.69	100%	1.69
	BPBreID + PSFP	185	0.8293 ± 0.006	0.104	8.3%	60%	0.72	81%	0.98
P-DukeMTMC-reID	Baseline	222	0.5254 ± 0.008	0.223	–	100%	1.90	100%	1.90
	BPBreID + PSFP	195	0.8102 ± 0.007	0.113	8.2%	65%	0.80	85%	1.05
Occluded-ReID	Baseline	230	0.4987 ± 0.009	0.241	–	100%	2.01	100%	2.01
	BPBreID + PSFP	200	0.7934 ± 0.008	0.121	8.4%	61%	0.77	82%	1.03

[†] Accuracy Drop (%) is relative to BPBreID (unpruned)

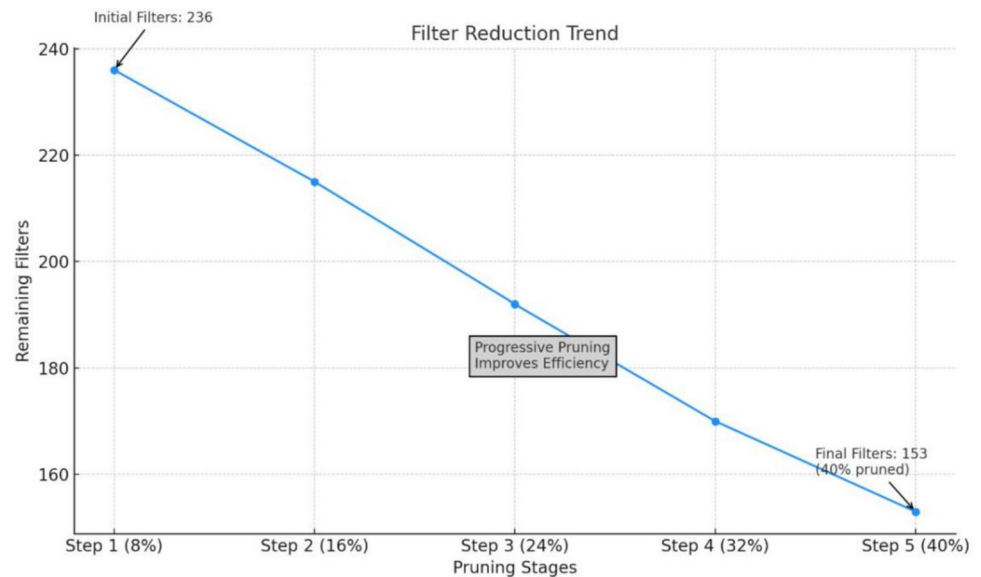
Lower FLOPs/Accuracy and Memory/Accuracy ratios indicate superior efficiency-performance balance

Bold values indicate the best-performing method for the corresponding evaluation metric

Table 5 Comparison with state-of-the-art pruning-based re-ID methods (with FPR & stability)

Method	Rank@1 ($\pm \sigma$)	mAP ($\pm \sigma$)	FPR ↓	Inference time (ms)	FLOPs reduction (%)
PRID (2021) [67]	0.840 $\pm$ 0.009	0.720 $\pm$ 0.008	0.118	210	25
PAT (2024) [65]	0.860 $\pm$ 0.007	0.740 $\pm$ 0.006	0.106	200	30
DFPC (2024) [66]	0.870 $\pm$ 0.006	0.755 $\pm$ 0.006	0.101	190	35
SFP (2024) [59]	0.855 $\pm$ 0.008	0.725 $\pm$ 0.007	0.111	205	27
BPBreID + PSFP (Proposed)	0.846 $\pm$ 0.005	0.678 $\pm$ 0.006	0.093	190	37

Bold values indicate the best-performing method for the corresponding evaluation metric

Fig. 4 Filter reduction trend across pruning stages, showing cumulative pruning from 8 to 40% and minimal accuracy loss due to fine-tuning after each stage

4. Figures 5 and 6 offer qualitative insights into its performance characteristics in multi-person and multi-camera settings, demonstrating successful identification with presence, overlap, and pose variance, thus providing further evidence to support its findings. It can thus be stated that with a total BPBreID model performance of 92.19% Rank-1 performance on Market-1501, there is a controlled loss in performance ($\sim 8.2\%$) but with significant efficiency gains. The proposed framework BPBreID + PSFP therefore offers better efficiency/accuracy tradeoffs, proving vital in real-time person Re-ID systems.

In Fig. 4, the trend of the filter reduction process among several levels of pruning within the process of compressing the model via Progressive Soft Filter Pruning is shown. In the diagram, the prune levels are represented on the x-axis and are marked with cumulative prune levels ranging from 8 to 40%, while the number of filters remaining for each prune level is represented on the y-axis. It is evident that the reduction in the total number of filters follows a linear reduction trend with the progressing prune process, implying the removal of redundant filters from the designed network.

“This controlled stepwise pruning is necessary in order to ensure stability in our model and to prevent sudden drops in performance that could happen when using more aggressive one-shot pruning strategies.” Through this act of pruning and fine-tuning its model, it is able to maintain a certain level of accuracy and distort only to a limited degree in the feature space, as exhibited in Tables 3 and 4. The above explanation gives a clear indication that through Fig. 4, it is able to validate its efficiency and accuracy trade-off, which is offered by its PSFP algorithm.

Figure 5, is a demonstration of the person re-identification technique through a search image matching people within a complex group scenario.

The image at the top left corner is the search image, while the bottom image is a real-world image of a crowded surveillance environment with people encircled within red rectangles, and the correct match is encircled within a green box.

Although encountering difficulties like diverse poses, background clutter, and occlusions, the model is able to

Fig. 5 Illustration of matching a query person image to corresponding individuals within a crowded group scene across camera views

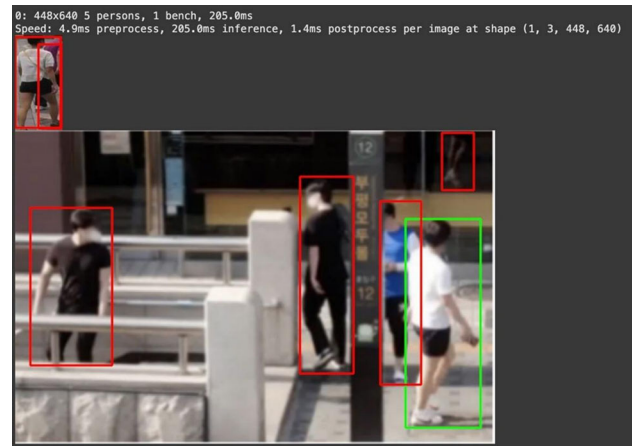
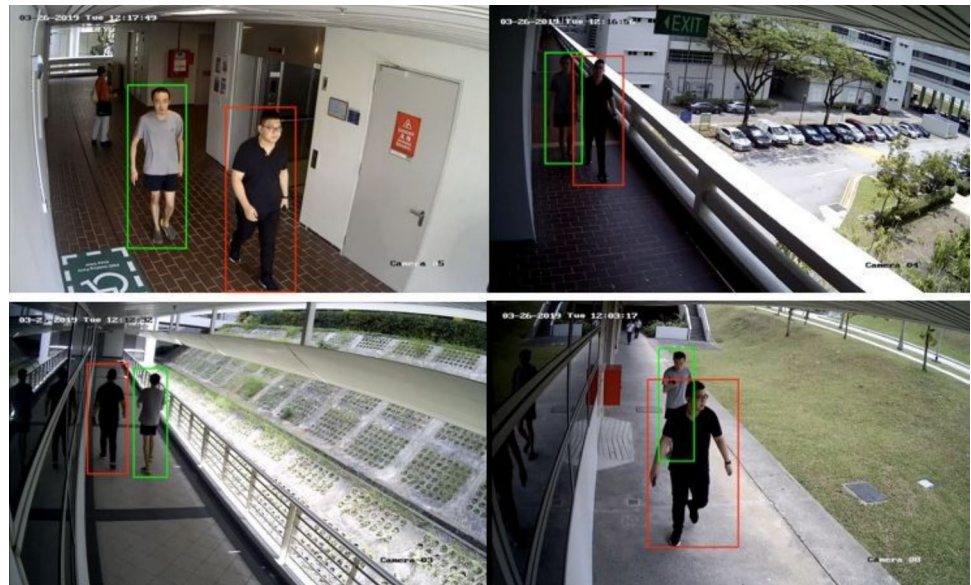


Fig. 6 Illustration of multi-camera person re-identification, showing the matching of the same individual across different locations and non-overlapping camera views



find the correct person by using global and localized characteristics of body parts. The following qualitative evaluation provides additional information to the quantitative evaluation of Rank-k, mAP, and occlusions presented in Sect. “[Results and discussion](#)”.

Figure 6 demonstrates multi-camera person re-identification across different surveillance locations. Bounding boxes indicate detected individuals across diverse viewpoints and lighting conditions. The proposed model successfully associates identities by learning invariant representations that remain stable across camera changes and partial occlusions.

These visual results reinforce the robustness claims supported by the occlusion-focused evaluation and feature-space consistency analysis (RRMSE) reported earlier, highlighting the effectiveness of BPBreID with PSFP in real-world cross-camera deployment scenarios.

Conclusion

This work presented an efficient and occlusion-robust person re-identification framework that integrates body part-aware global–local feature learning with Progressive Soft Filter Pruning (PSFP). By combining visibility-aware body part representations with structured, stage-wise filter pruning, the proposed BPBreID + PSFP model achieves a favorable balance between recognition performance and computational efficiency. Extensive experiments on multiple benchmark datasets demonstrate that the unpruned BPBreID model achieves strong Rank-1 and mAP performance, while the pruned variant retains over 91% of the original Rank-1 accuracy with up to 37% reduction in FLOPs and notable memory savings. Feature-space analysis further confirms that PSFP preserves discriminative identity representations despite aggressive compression.

Despite these promising results, several limitations remain. First, the efficiency gains achieved through PSFP

come with a consistent performance trade-off of approximately 8% in Rank-1 accuracy, highlighting the inherent tension between compactness and discriminative power. Second, the evaluation is limited to image-based, closed-set Re-ID benchmarks, which may not fully capture the challenges posed by real-world deployments involving open-set identities, domain shifts, or continuous video streams. Additionally, pruning decisions are static after training and do not adapt to input-specific factors such as occlusion severity or visual ambiguity.

Future work will focus on addressing these limitations along multiple directions. One promising avenue is the exploration of dynamic or input-adaptive pruning strategies that adjust network capacity based on scene complexity or occlusion level at inference time. Integrating uncertainty-aware modeling could further improve reliability by enabling confidence estimation in ambiguous or heavily occluded scenarios. Extending the framework to video-based person Re-ID is another important direction, where temporal cues can complement spatial body-part representations. Finally, deployment-oriented evaluations on edge platforms, incorporating latency, power consumption, and privacy-aware learning mechanisms, will be essential to validate the practical applicability of the proposed approach.

Funding Open access funding provided by Datta Meghe Institute of Higher Education and Research. This research received no specific grant from any funding agency, commercial entity, or not-for-profit organization.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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